

Henry Tejada

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Education

Franklin W. Olin College of Engineering | Needham, MA | GPA: 3.67

Bachelor's of Science in Electrical and Computer Engineering

Coursework: Principles of Integrated Engineering, Computer Architecture, Circuits, Software Design, Engineering Design, Modeling and Simulation

Lone Star College- North Harris | Houston, TX | GPA: 3.96

Associate's of Science in General Math and Science

Spring Early College Academy | Houston, TX | GPA: 3.95

High School Diploma

Relevant Skills

- KiCad Circuit Design
- MATLAB
- LT Spice
- PCB Soldering
- SolidWorks
- Python
- Microsoft Office

Projects

- **Disk Golf Disk Cleaner** - Created an enclosure and support structure using Onshape and Fusion to hold a spinning disk and upright-facing motor. Manufactured various parts out of wood using a laser cutter and drill press.
- **STM32 Transition Development PCB** - Designed a PCB using KiCad for the hardware and firmware development for Olin Electric Motorsports.
- **Brakes Board Modifications** - Modified a circuit board's schematics using KiCad that monitors a car's motor current and brake pedal status, for safety reasons, to improve reliability.
- **Texture Measurement Device** - Created a circuit that measured the roughness of a surface based on the scattering of light emitted from the circuit.
- **RV32I RISC-V Processor** - Implemented an Iverilog-compatible implementation of a RISC-V processor with most RV32I instructions.
- **Tanks Game Replica** - Created a replica of the popular Tanks game for Nintendo Switch using Python and the PyGame library.
- **Line Following Robot** - Designed mounts for auxiliary parts for the robot using SolidWorks
- **Hopper Project** - Designed a device that hops about 2 mm, imitating squid jet propulsion's motion using SolidWorks.

Experience

Electrical Engineering Co-op | TTM Technologies - Advanced Manufacturing Group | Stafford Springs, CT | May 2026 - August 2026

- Created schematics from an existing 4-layer PCB and speced out necessary components to remake the same functionality.
- Cross-sectioned and imaged defects in microvia coupons for a study on the reliability of a machine.
- Reverse engineered firmware from a dsPIC and PIC microcontroller using Ghidra and MPLAB software.
- Communicated with manufacturers to procure missing components.

Process Engineering Co-op | TTM Technologies - Advanced Manufacturing Group | Stafford Springs, CT | May 2025 - August 2025

- Refactored alert system module to account for a new data paradigm and to allow for the future implementation of advanced analytics using Ignition.
- Pioneered standardization of coding practices and design styles by creating documentation to develop an easily maintainable and scalable platform.
- Designed unified tools for the updated system to speed up development and allow for easy updates of the entire system should there be changes to the data acquisition system.
- Implemented a user-friendly UI in Ignition, forming a cohesive and uniform style across the different webpages in different modules of the platform.
- Implemented an updated data structure, making use of user-defined types, to standardize data acquisition across all TTM PCB fab sites.
- Assisted in running a factory acceptance test for a machine that was being built for a customer.

Co-Electrical Subsystem Lead | Olin Electric Motorsports | Needham, MA | May 2025 - Present

- Implemented an updated THT and SMT parts organization system that has reduced search time for parts by 50%.
- Procured parts used by custom PCBs from various vendors, sourcing the most economical option that meets our requirements.
- Aided in the ATmega 16M1/64M1 -> STM32 hardware transition, specifically by developing a development PCB for testing and sourcing new parts needed for the new microcontroller.
- Helped new members learn more about our electrical system and taught new members about circuits and harnessing.
- Created a timeline for the electrical subsystem, accounting for our two-year design cycle, 1 semester of debugging, and the mechanical subsystem timeline.

Low Voltage Subteam Member | Olin Electric Motorsports | Needham, MA | September 2024 - May 2025

- Updated circuitry in the brake system and front motherboard custom PCBs to account for changes in the 2025 FSAE Competition rules using KiCad.
- Updated firmware coded in C to account for changes in circuitry and to allow for more hardware diagnostics.
- Acted as a liaison between the mechanical subsystem and electrical subsystem to ensure the proper space was allocated for the front motherboard.
- Documented changes and design decisions made to the brake system and front motherboard custom PCBs to allow for knowledge transfer between team members.
- Created a Raspberry Pi setup that allows the user to use SSH to remotely access the Raspberry Pi, and for the user to know when the Raspberry Pi is ready to be remotely accessed.

Utility | Rebecca's Culinary Group | Needham, MA | September 2024 - Present

- Assisted coworkers with various tasks, including preparation, cleaning, and maintaining a good presentation of the serving area.